

REVIT CONCRETE MODELING

20 Sections in this course outline

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SECTIONS

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01 Brief overview of Revit (what it entails)

- ✓ Difference with AutoCAD software
- ✓ Benefits over AutoCAD software

03 Beginning with Revit

- ✓ Creating a new project
- ✓ An overview of the templates
- ✓ An overview of palettes, tabs and panels

05 Brief Overview of Creating 3D elements

- ✓ Foundations, Columns, Beams, Slabs

02 Setting up the software

- ✓ Brief introduction of Libraries required
- ✓ Downloading other extensions from Autodesk site

04 Creating Datum Elements

- ✓ Creating Levels
- ✓ Creating grids – (same as with the levels creation)
- ✓ Propagate options extends on the grid
- ✓ Customizing and changing the shape and appearance of grid bubbles
- ✓ Linking CAD files to create grids

06 Creating Columns

- ✓ Placement of columns
- ✓ Loading column families



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07 Creating Foundations

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- ✓ Hosting foundations on the columns created
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- ✓ Using eccentricity option
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- ✓ Loading foundation families into your drawings

09 Curved Beams

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- ✓ Procedure of modeling the curved beams

08 Creating Beams and slabs

- ✓ Copying beams and slabs from level to level
- ✓ Creating a slab
- ✓ Hollow pot and waffle slab

11 Creating Staircase and Ramps

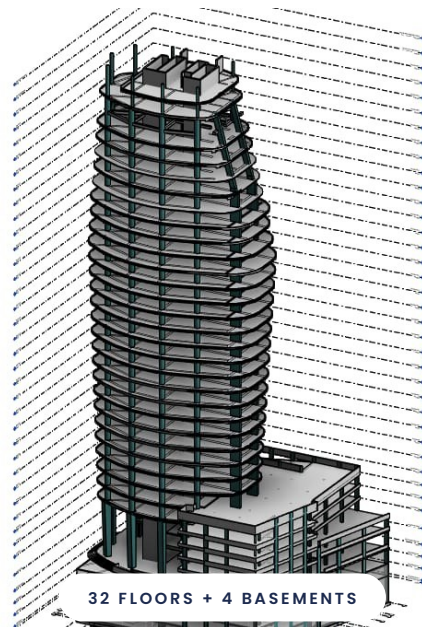
- ✓ Types of staircase
- ✓ Creating an architectural ramp and structural ramp
- ✓ Modeling Staircase as a ramp
- ✓ Shaft openings

10 Beam System Layout and Justification Rules

- ✓ Using the beam system to model the structural beams
- ✓ Modelling the beams using the beam system layout rules
- ✓ Modelling the beams using the beam system Justification Rules
- ✓ Deselecting the beam system

12 Visibility/Graphics and Filters

- ✓ Applying Visibility/Graphic Overrides — Method 1
- ✓ Applying Visibility/Graphic Overrides — Method 2
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13 Creating Roofs and Truss placement

- ✓ Sloped and gable ended roofing
- ✓ Adding system trusses on your roof
- ✓ Editing and creating your own customized trusses

15 Annotations

- ✓ Adding dimensions
- ✓ Adding spot elevations
- ✓ Adding spot coordinates
- ✓ Adding spot slope

17 Schedules and Material Takeoffs

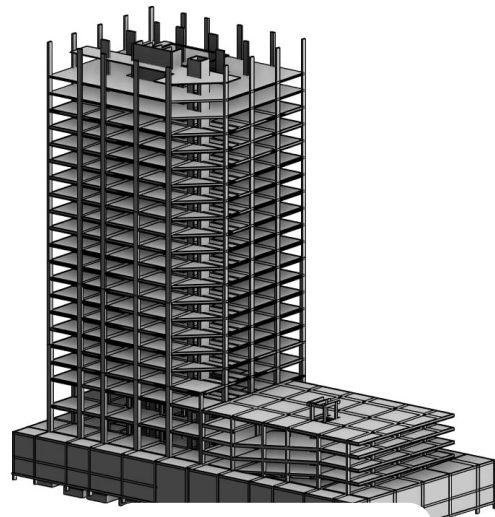
- ✓ Creating Bar Bending Schedules
- ✓ Loading shape images into bar bending schedules
- ✓ Creating Material Takeoffs for the structural elements
- ✓ Sorting and filtering fields in material takeoffs and bar bending schedules

14 Placing of Reinforcement

- ✓ Creating sections in your drawing
- ✓ Foundation Reinforcement
- ✓ Column Reinforcement
- ✓ Beam Reinforcement
- ✓ Slab reinforcement
- ✓ Staircase reinforcement
- ✓ Ramp reinforcement

16 Detailing of Sections

- ✓ Placing breaklines in the sections
- ✓ Using hatch files to create regions
- ✓ Placement of rebar tags on your reinforcements
- ✓ Adding comments on your reinforcement
- ✓ Tagging of the structural elements





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18 Placing Drawings on Sheets

- ✓ Loading different paper sizes for your title blocks
- ✓ Using system loaded title blocks
- ✓ Editing and customizing the title blocks
- ✓ Adjusting texts and fields in your title block
- ✓ Saving title blocks as a Revit family template
- ✓ Introduction to duplicate options for drawings
- ✓ Placing drawings on the sheets

19 Printing Sheets to PDF format

- ✓ Setting up your sheets before printing
- ✓ Printing one sheet at a time
- ✓ Printing several sheets at a time

20 Conversion of your files into CAD and other formats

- ✓ Export options for your drawing into other formats
- ✓ Transporting your drawing to AutoCAD
- ✓ Sending your model for analysis

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Revit Concrete Structure Training

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